

Crime Analysis Using Tableau

1. Introduction

Crime analysis plays a crucial role in maintaining law and order by helping law enforcement agencies identify crime patterns, trends, and high-risk locations. Through effective analysis of historical crime records, police departments can allocate resources efficiently, improve public safety, and develop crime prevention strategies.

This project uses Tableau to analyse crime data and create interactive dashboards that provide insights into crime distribution, temporal trends, geographical hotspots, and comparative crime characteristics. The visualizations help transform raw crime data into actionable information for decision-making.

2. Objectives

The main objectives of this project are:

- Analyse the overall crime statistics and identify the most common crime types.
- Identify geographical crime hotspots across districts.
- Examine crime patterns based on day of week, hour, and time blocks.
- Analyse crime trends over multiple years.
- Compare crime incidents based on arrest status and domestic crime involvement.
- Develop interactive Tableau dashboards for effective visualization and reporting.

3. Dataset Description

The dataset contains historical crime records with information related to:

- Case Number
- Date
- Primary Crime Type
- Location Description
- District
- Arrest Status
- Domestic Crime Indicator
- Latitude and Longitude
- Community Area
- Ward
- Year

The dataset consists of 247,797 crime records used for analysis and visualization.

4. Methodology

The project was completed using Tableau Desktop following the steps below:

Data Preparation

- Imported the crime dataset into Tableau.
- Verified data types and field formats.
- Created calculated fields for:
 - Day of Week
 - Hour
 - Time Block (Morning, Afternoon, Evening, Night)

Data Visualization

Four interactive dashboards were developed:

1. Overall Crime Statistics Dashboard
2. Time Period Analysis Dashboard
3. Trend Analysis Dashboard
4. Comparative Analysis Dashboard

5. Dashboard 1 – Overall Crime Statistics

Overview

This dashboard provides a summary of overall crime activity and geographical crime distribution.

Visualizations Included

- Total Crimes KPI
- Most Common Crime Types
- Crime Distribution by District
- Recent Crime Reports

Key Findings

- Total reported crimes: 247,797
- Theft was the most frequently reported crime.
- Battery and Criminal Damage were the next most common crime categories.
- Crime incidents were concentrated in specific districts, indicating potential crime hotspots.

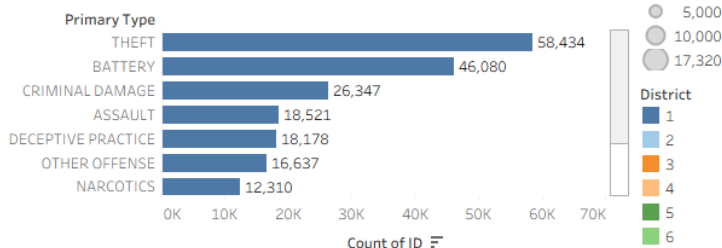
Overall Crime Statistics Dashboard

KPI - Total Crimes

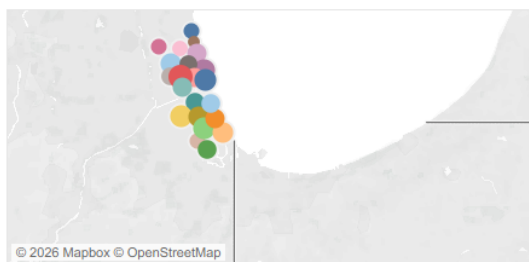
TOTAL CRIMES

2,47,797

Most Common Crime Types



Crime Distribution by District



Recent Crime Reports

Year of Date	Primary Type	District	Location Description	
2017	ARSON	1	CONSTRUCTION SITE	Ab
		2	ALLEY	Ab
			APARTMENT	Ab
			SCHOOL, PUBLIC, BUILDIN..	Ab
		3	APARTMENT	Ab
			RESIDENCE	Ab
			RESIDENTIAL YARD (FRO..	Ab
			STREET	Ab
		4	PARK PROPERTY	Ab
				Ab

6. Dashboard 2 – Time Period Analysis

Overview

This dashboard examines crime occurrence based on time-related factors.

Visualizations Included

- Crimes by Day of Week
- Crimes by Hour
- Crime Incidents by Time Block

Key Findings

- Crime incidents were relatively evenly distributed throughout the week.
- Crime activity was lowest during early morning hours.
- Crime levels increased during daytime and evening periods.
- Night and Afternoon periods accounted for the highest percentage of reported crimes.

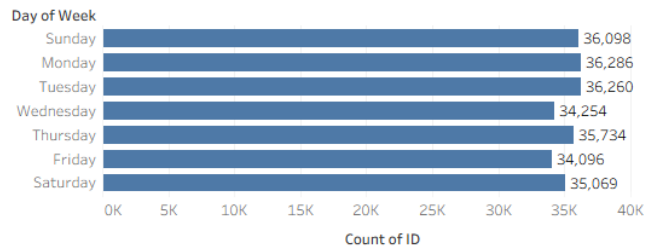
Time Block Distribution

Time Block Percentage

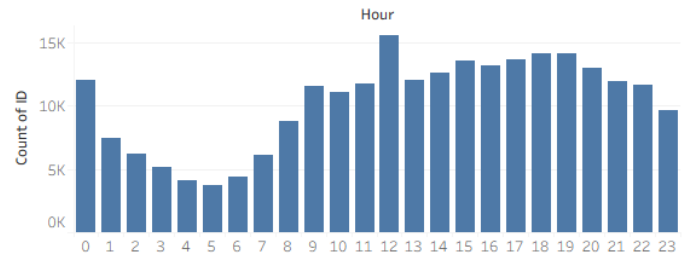
Night	27.61%
Afternoon	27.04%
Morning	23.18%
Evening	22.18%

Crime Analysis Dashboard – Time Period Analysis

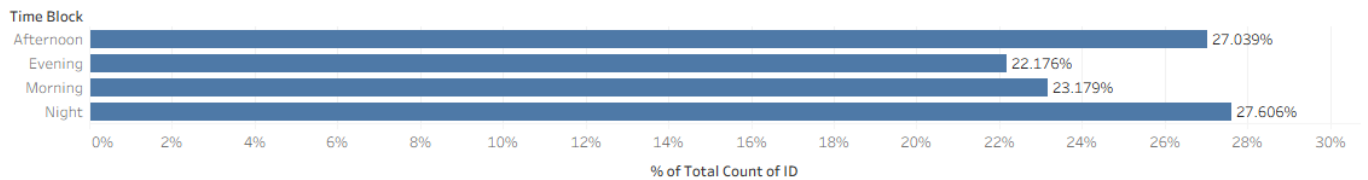
Crimes by Day of Week



Crimes by Hour



Crime Incidents by Time Block



7. Dashboard 3 – Trend Analysis

Overview

This dashboard analyses crime trends over multiple years.

Visualizations Included

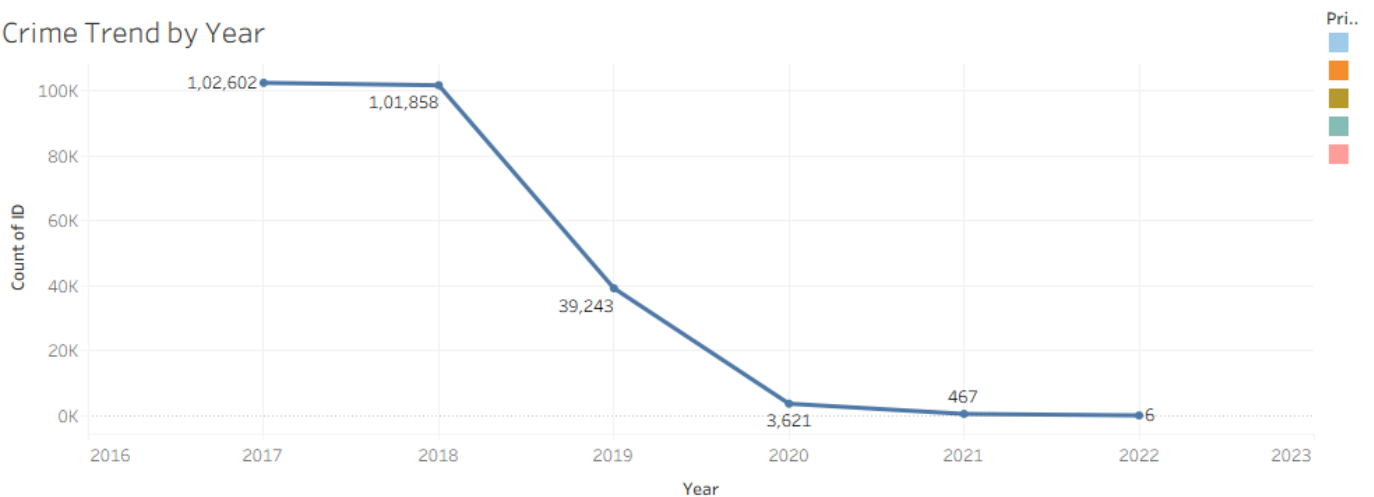
- Crime Trend by Year
- Crime Type Trend by Year

Key Findings

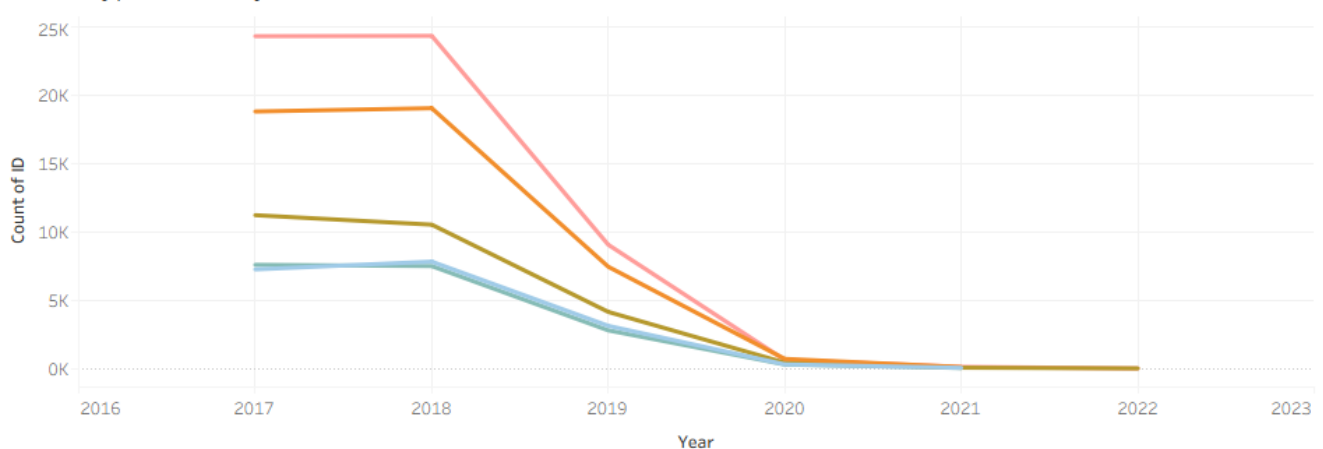
- Crime incidents were highest during 2017 and 2018.
- A significant decline was observed after 2019.
- Theft consistently remained one of the most common crime categories.
- Battery and Criminal Damage also showed high reporting frequencies across the analysed years.

Crime Analysis Dashboard – Trend Analysis

Crime Trend by Year



Crime Type Trend by Year



8. Dashboard 4 – Comparative Analysis

Overview

This dashboard compares crime incidents using different categorical factors.

Visualizations Included

- Arrest Analysis
- Domestic Crime Analysis
- Crime Type vs Arrest

Key Findings

Arrest Analysis

- Crimes resulting in arrest: 50,253
- Crimes without arrest: 197,544

Domestic Crime Analysis

- Domestic crimes: 41,018

- Non-domestic crimes: 206,779

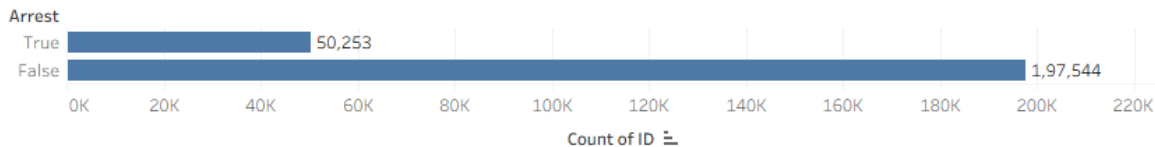
Crime Type Comparison

- Arrest rates varied across crime categories.
- Certain crime types showed substantially lower arrest rates compared to reported incidents.

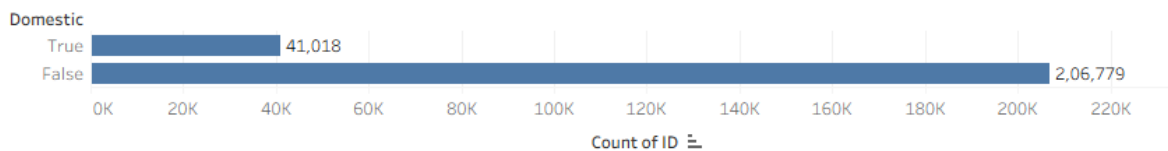
Crime Analysis Dashboard – Comparative Analysis

Arrest Analysis

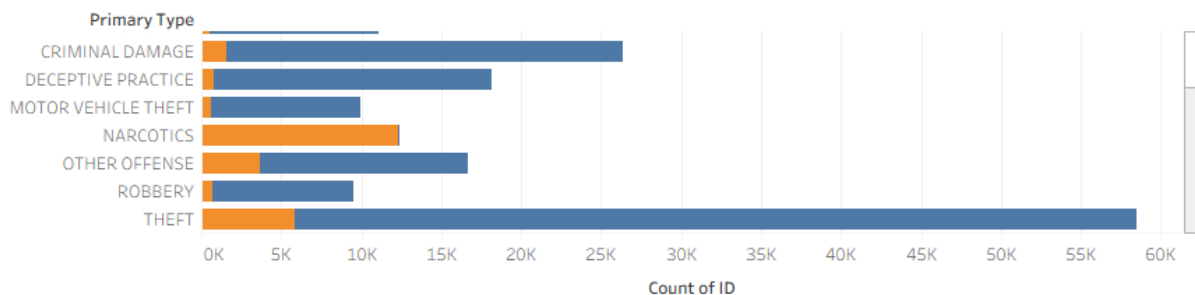
Arrest
False
True



Domestic Crime Analysis



Crime Type vs Arrest



9. Overall Findings

The analysis revealed several important insights:

- Theft is the most frequently reported crime type.
- Crime activity is distributed across multiple districts, with some districts experiencing significantly higher crime volumes.
- Crime incidents occur throughout the day but are slightly higher during afternoon and night periods.
- Crime rates showed a decreasing trend in later years of the dataset.
- Most reported crimes did not result in an arrest.
- Non-domestic crimes significantly outnumber domestic crimes.

These findings can assist law enforcement agencies in identifying priority areas for resource allocation and crime prevention initiatives.

10. Conclusion

This project successfully utilized Tableau to analyse crime data and develop interactive dashboards for crime monitoring and decision-making. Through geographical, temporal, trend, and comparative analyses, meaningful insights were generated regarding crime patterns and reporting behaviour.

The dashboards provide a user-friendly and interactive platform for exploring crime statistics, enabling better understanding of crime trends and supporting data-driven law enforcement strategies.